An ILP for joint binarization, cell ordering and signed threshold network inference from single-cell snapshots

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This note extends the STOP-BIT program of Section 3.3 of the manuscript (*Inferring Synchronous Boolean Models From Noisy Time-Series Data*) to single-cell snapshot data with no observed time order, such as the Hamey et al. haematopoietic stem and progenitor cell qRT-PCR data. The function representation (Sections 3.3.1–3.3.2), the agreement indicators (Section 3.3.3) and the edge penalties (Section 3.3.4) are reused unchanged. What is new is that each cell has one latent state rather than a trajectory row, and that the program chooses a predecessor for every cell, so that the cells form a tree whose edges are model transitions.

1 Data and notation

Genes are $V = [n]$ and cells are $[N]$. The observed data are $\tilde{X} \in \{0,1\}^{N \times n}$, with $\tilde{X}_{c,j} = 1$ iff gene j was detected in cell c (raw $C_t < 40$), the binarization used by Hamey et al. Let $R \subseteq V$ be the genes allowed to act as regulators (Section 7). The candidate inputs of node j are $I_j = R \cup \{j\}$: every regulator, and j itself for a self-activation. No prior-knowledge network (PKN) is used, i.e. the manuscript's $\tilde{E}$ is empty. Its α term then vanishes, and since the normalizer $|\tilde{E}|$ is zero we charge each edge $\beta \ln N$ changed data values instead, a cost that grows with the number of cells as in BIC. If a PKN is used (Section 7), the two edge terms of Section 3.3.4 apply unchanged. The stability weight is written γ to avoid a clash with the manuscript's α, and d bounds the in-degree as in constraint (2) of the manuscript.

We add a *pseudo-root*, index 0: a latent state that is not a cell and has no data row, and that is the predecessor of every cell at pseudotime 1. For every cell c let $\mathrm{NN}_k(c)$ be the k cells nearest to c in Hamming distance on $\tilde{X}$ (ties broken by index), and let its candidate predecessor cells be the symmetrized neighbourhood

$$P(c) = \{a \neq c \mid a \in \mathrm{NN}_k(c) \text{ or } c \in \mathrm{NN}_k(a)\}.$$

Symmetrizing makes $a \in P(c)$ iff $c \in P(a)$, so the candidate set favours neither direction of an edge and orientation is left to the program. Let $H \in \mathbb{N}$ be the maximal number of steps on any path from the pseudo-root; $H = N$ makes it non-binding.

Why restrict predecessors. Without restriction the predecessor variables and transition constraints grow as N^2 and $N^2 n$, about 4×10^8 constraints for all cells. A distant predecessor a of c is only usable if a is unstable on every gene where they differ, which the stability penalty makes costly anyway, so few optimal edges are lost. The neighbourhood is computed on the data $\tilde{X}$, not on the latent states, so k should leave

room for latent flips: $k = 20$ gives on average $|P(c)| = 31$ candidates on the full data (maximum 149). The pseudo-root is a candidate for every cell, so the restriction never makes the program infeasible.

Variables.

- $Y \in \{0,1\}^{(N+1)\times n}$: the latent (denoised) state $Y_{c,*}$ of each cell $c \in [N]$, and the pseudo-root state $Y_{0,*}$.
- θ, p, q, u: thresholds, activator and repressor indicators, and input-node indicators, exactly as in Sections 3.3.1–3.3.2, with $\Sigma_{i,j} = p_{i,j} - q_{i,j}$.
- $F(Y_{c,*})_j \in \{0,1\}$: the model's update of gene j applied to state $c \in \{0\} \cup [N]$.
- $C_{c,j} \in \{0,1\}$: agreement of $Y_{c,j}$ with $\tilde{X}_{c,j}$.
- $U_{c,j} \in \{0,1\}$: instability of gene j at state $c \in \{0\} \cup [N]$, $U_{c,j} = 1$ iff $F(Y_{c,*})_j \neq Y_{c,j}$.
- $\pi_{a,c} \in \{0,1\}$ for $a \in \{0\} \cup P(c)$: a is the predecessor of c.
- $\tau_c \in \{1, \ldots, H\}$: the discrete pseudotime of cell c, with the constant $\tau_0 = 0$.

2 Subprograms

Functions. Denote by FUNCTIONS(F, Y, θ, p, q, u) the subprogram of Sections 3.3.1–3.3.2, applied once to every state: for each $c \in \{0\} \cup [N]$, the auxiliary indicators $a(Y_{c,*})_{i,j}$ for $i \in I_j$ with constraints (3), the big-M output constraints (4) as modified for input nodes, and the input-node constraints, defining $F(Y_{c,*})_j$. In the manuscript the same subprogram also sets $Y^l_{t+1,*} = F(Y^l_{t,*})$ along each time series; here it only defines the outputs, and the link between cells is made by the transition constraints below. We additionally forbid negative self-loops,

$$q_{j,j} = 0 \qquad \forall j \in V, \tag{1}$$

since $f_j(x) = 1 - x_j$ makes every gene unstable at every state and so explains every change (Hamey et al.'s PKN contains self-activations only).

Agreement. Unchanged from Section 3.3.3, with one data row per cell:

$$C_{c,j} \leq 1 + Y_{c,j} - \tilde{X}_{c,j}, \qquad C_{c,j} \leq 1 - Y_{c,j} + \tilde{X}_{c,j} \qquad \forall c, j. \tag{2}$$

This is the binarization penalty: a latent state may differ from the detection call at a cost per bit.

Instability. INSTABILITY(U, F, Y) makes $U_{c,j}$ the exclusive or of $F(Y_{c,*})_j$ and $Y_{c,j}$, for every $c \in \{0\} \cup [N]$:

$$\begin{aligned} &U_{c,j} \geq F(Y_{c,*})_j - Y_{c,j}, && U_{c,j} \geq Y_{c,j} - F(Y_{c,*})_j, \\ &U_{c,j} \leq F(Y_{c,*})_j + Y_{c,j}, && U_{c,j} \leq 2 - F(Y_{c,*})_j - Y_{c,j} \qquad \forall c \in \{0\} \cup [N],\ j. \end{aligned} \tag{3}$$

Both directions are needed even though U is penalized: U also relaxes the transition constraints, so a one-sided bound would let the solver buy a change the model does not make.

Tree. TREE(π, τ) gives each cell exactly one predecessor and orders cells by pseudotime:

$$\pi_{0,c} + \sum_{a \in P(c)} \pi_{a,c} = 1 \qquad \forall c, \tag{4}$$

$$\tau_c \geq \tau_a + 1 - H(1 - \pi_{a,c}) \qquad \forall c,\ a \in P(c), \tag{5}$$

$$\tau_c \leq 1 + (H-1)(1 - \pi_{0,c}) \qquad \forall c. \tag{6}$$

By (5) a cell predecessor has a strictly smaller pseudotime, so following predecessors from any cell ends at the pseudo-root and the chosen edges form a tree on $\{0\} \cup [N]$ of depth at most H. A cell with $\tau_c = 1$ cannot have a cell predecessor, and by (6) every child of the pseudo-root has $\tau_c = 1$; so the cells at pseudotime 1 are exactly the children of the pseudo-root. When $\pi_{a,c} = 0$, (5) reads $\tau_c \geq \tau_a + 1 - H$, which holds since $\tau_a \leq H$ and $\tau_c \geq 1$.

Transitions. TRANSITIONS(π, Y, U): if a is the predecessor of c, then c may differ from a only on genes unstable at a,

$$Y_{c,j} - Y_{a,j} \leq U_{a,j} + 1 - \pi_{a,c}, \qquad Y_{a,j} - Y_{c,j} \leq U_{a,j} + 1 - \pi_{a,c} \qquad \forall c,\ a \in \{0\} \cup P(c),\ j \in V. \tag{7}$$

For binary values, a gene unstable at a has $F(Y_{a,*})_j = 1 - Y_{a,j}$, so on every gene c either keeps a's value or takes the model's update: $Y_{c,j} \in \{Y_{a,j}, F(Y_{a,*})_j\}$. This is a one-step transition in which any subset of genes updates, a relaxation of the synchronous scheme of Section 2.1 suited to snapshots of cells whose genes need not switch in lockstep. The synchronous variant replaces (7) with $|Y_{c,j} - F(Y_{a,*})_j| \leq 1 - \pi_{a,c}$. The pseudo-root is a predecessor like any cell, so its children lie between $Y_{0,*}$ and $F(Y_{0,*})$: a single learned origin state for all initial cells.

3 The full program

$$\begin{aligned}
\text{maximize} \quad & \frac{1}{Nn} \sum_{c,j} C_{c,j} - \frac{\beta \ln N}{Nn} \sum_{j} \sum_{i \in I_j} (p_{i,j} + q_{i,j}) - \frac{\gamma}{Nn} \sum_{c,j} U_{c,j} - \frac{\rho}{N} \sum_{c} \pi_{0,c} \\
\text{subject to} \quad & \text{FUNCTIONS}(F, Y, \theta, p, q, u), \ (1) \\
& (2) \\
& \text{INSTABILITY}(U, F, Y) \\
& \text{TREE}(\pi, \tau) \\
& \text{TRANSITIONS}(\pi, Y, U).
\end{aligned}$$

The first two terms are the STOP-BIT objective with cells in place of time-series rows and no PKN. The third is the stability regularization, $\gamma \sum_c \text{dist}(F(Y_{c,*}), Y_{c,*})$ with dist the Hamming distance, normalized like the agreement term; it includes the pseudo-root. The fourth charges ρ/N for each cell placed under the pseudo-root. It is needed even though the pseudo-root's children must satisfy the transition constraints: $Y_{0,*}$ has no data row, so the model can make it unstable on every gene ($F(Y_{0,*}) = 1 - Y_{0,*}$), at a total cost of only $n\gamma/(Nn) = \gamma/N$. The pseudo-root is then a parent of any state, and without ρ every cell could be placed under it, removing all other transitions.

Balancing the weights. All terms are on a per-bit scale: one latent bit that disagrees with the data costs $1/(Nn)$, one unstable gene at one cell costs $\gamma/(Nn)$, and one root costs $\rho/N = \rho n/(Nn)$.

- Roots. A cell whose nearest candidate predecessor (in latent state) differs on h genes can always be attached by flipping those h bits, at cost $h/(Nn)$. It is therefore placed under the pseudo-root only if $\rho n < h$ and no cheaper explanation exists: ρn is the number of unexplained differences a root is worth. On the full data the mean nearest-neighbour distance is 3.2 of 42 genes, so for example $\rho n = 6$ makes roots of cells more than six flips from any other.

- Stability (see also Section 4). A change on gene j from a to a child can be paid for by one flip of a latent bit, $1/(Nn)$ per child, or by making j unstable at a, $\gamma/(Nn)$ once for all m children of a that change on j. Instability is preferred when $\gamma < m$. A tree on $N + 1$ states has on average about one child per state, so the useful range is $\gamma \in (0, 1)$. Once γ exceeds the typical number of children sharing a change, differences are paid by flips, the model goes unused, and cells collapse into few distinct states. As $\gamma \to 0$ the model can be unstable everywhere and explain anything, limited only by consistency of F across cells and the edge penalties; constraint (1) blocks the simplest such model.

- Edges. As in Appendix .4 of the manuscript, an edge is worth adding when it lets the model explain $\beta \ln N$ more bits (through fewer flips or fewer unstable genes). An earlier version charged βN bits per edge; the benchmark results below with $\beta = 0.1$, 0.01 and 0.001 used it.

Tree shape. No penalty on the width or height of the tree is added. Height is free: a chain of identical or near-identical cells costs nothing, which is the intended reading of densely sampled states, and H only needs to be large enough not to bind. Width is already shaped by the objective: an unstable gene at a is paid once and explains that change in every child, so the stability term favours parents with several children changing in the same way, which is the branching we want to recover. The risk is a hub, one parent with many unstable genes adopting many unrelated children. The neighbourhood $P(c)$ limits this, since a hub can only adopt cells near it; if hubs still appear, a cap $\sum_c \pi_{a,c} \le D$ on the number of children is a simple check.

4 Choosing γ

Flips and instabilities are both counted per (state, gene) bit, so γ is the price of one unstable gene at one state in flipped bits. A principled value comes from reading the objective as a negative log-posterior, under two assumptions:

- Each observed bit is the latent bit flipped independently with probability ε, i.e. $\tilde{X}_{c,j} = Y_{c,j}$ with probability $1 - \varepsilon$.

- As a prior on the network, each (state, gene) is unstable independently with probability π_U.

Up to constants, the negative log-posterior is then

$$\log \frac{1-\varepsilon}{\varepsilon} \sum_{c,j} [Y_{c,j} \neq \tilde{X}_{c,j}] + \log \frac{1-\pi_U}{\pi_U} \sum_{c,j} U_{c,j},$$

and dividing by the first coefficient gives one flip a cost of 1 and

$$\gamma = \frac{\log\big((1-\pi_U)/\pi_U\big)}{\log\big((1-\varepsilon)/\varepsilon\big)}.$$

So $\gamma < 1$ exactly when instability is a priori more common than measurement error ($\pi_U > \varepsilon$). The prior is a heuristic: U is determined by F and Y, not drawn independently.

Both rates can be estimated from nearest-neighbour distances, but not from distances between arbitrary pairs. Arbitrary cells are separated by many differentiation steps (8.3 of 31 genes on average on the LMPP branch), so that distance measures global heterogeneity, not one step. Nearest neighbours bound what happens on a single tree edge: their mean distance h (2.2 of $n = 31$ on the LMPP branch) is roughly the noise from both cells, $2\varepsilon n$, plus the true changes per edge, about $\pi_U n$ if every unstable gene updates. So $\varepsilon \le h/(2n)$, and

$$\pi_U \approx \frac{h}{n} - 2\varepsilon.$$

For example $\varepsilon = 0.02$ gives $\pi_U \approx 0.031$ and $\gamma \approx 0.88$, and $\varepsilon = 0.01$ gives $\pi_U \approx 0.051$ and $\gamma \approx 0.64$. ε is the weakly determined quantity; held-out entry masking (hide some bits, score their recovery) can then choose γ around this value. The root penalty can be set the same way, $\rho n = \log\big((1-\pi_R)/\pi_R\big)/\log\big((1-\varepsilon)/\varepsilon\big)$ for a prior root probability π_R per cell.

5 Benchmark: START-BIT on candidate pairs

`snapshot_tree_heuristic.py` solves the program without a solver, as a benchmark and as a MIP start:

1. Every ordered pair (a, c) with $a \in P(c)$ is a transition from $\tilde{X}_{a,*}$ to $\tilde{X}_{c,*}$. Both directions are included, since the orientation is unknown.
2. START-BIT (Section 3.2 of the manuscript) fits each node's signed threshold function to these transitions, with an edge cost of βT agreed cells for T transitions, the same fraction β/n of all scored cells as in the objective. A fit with a negative self-loop is refitted without the node's own input.
3. With F fixed and $Y = \tilde{X}$, edge $a \to c$ costs the number of genes on which c differs from a while F leaves them unchanged at a, and a root edge costs ρn. The tree is the minimum-cost spanning arborescence rooted at the pseudo-root (Edmonds' algorithm).
4. Going down the tree, each child's violating bits are set to its parent's values, which makes the solution feasible.

The table below is round 1 of the alternation described after it.

Effective stability weight. For gene j, a pair (a, c) on which j does not change is scored correct iff $F(\tilde{X}_{a,*})_j = \tilde{X}_{a,j}$, i.e. iff j is stable at a. The all-pairs score of node j is therefore the number of changing pairs it explains plus a stability bonus at each state a, weighted by the number of a's candidate neighbours that agree with a on j. On the LMPP branch ($k = 20$) that is 26.5 on average per (state, gene), against 3.9 changing pairs. The all-pairs fit thus behaves like $\gamma \approx 26$, far above the values of Section 4: the fitted networks leave every state stable and explain no changes. The option `--changes-only` fits node j only on the pairs where j changes, which is $\gamma = 0$: the fit then pushes F_j to flip j at almost every state. Matching the program requires counting each state's stability once, with weight γ, which needs per-sample weights in START-BIT.

Results on the LMPP branch ($k = 20$, $\gamma = 0.5$, $\rho n = 6$; objective of the program). Collapsing every cell onto one state in a single chain scores about 0.81.

fit	β	objective	agreement	edges	unstable bits
all pairs	0.1	0.780	0.810	10	0
all pairs	0.01	0.787	0.800	42	0
all pairs	0.001	0.772	0.782	244	136
changes only	0.1	0.711	0.755	12	667
changes only	0.01	0.633	0.802	37	7,110
changes only	0.001	0.618	0.878	347	11,245

Alternation. Round 1 is steps 1–4. Each later round refits F to the current tree's edges (parent to child, on $\tilde{X}$; edges from the pseudo-root are left out) and rebuilds the tree for the new F. It runs for up to L rounds (default $L = 10$), stops when a round reproduces the previous tree, and returns the best round. On the LMPP branch:

- $\beta = 0.1$: a fixed point at round 6, best objective 0.784 (round 3). The network stays stable at every state throughout (8–10 edges, no unstable bits), so the result is a collapse onto the pseudo-root's state. A tree has only $N - 1 = 530$ transitions and each gene changes on few of them, while an edge costs $\beta T \approx 53$ agreed cells, so no edge that makes a state unstable pays for itself. The same holds in the program: an edge must save $\beta N \approx 53$ flipped bits.
- $\beta = 0.01$ and 0.001: no convergence in 10 rounds. About 200–280 of the 531 parents change every round, and the objective oscillates (0.753–0.791 and 0.625–0.772), with best objectives 0.791 (round 4) and 0.772 (round 1).

Every variant stays below the collapse, about 0.81. Raising L would not help: the two steps optimize different scores. The tree step minimizes the program's flips given F, but the refit maximizes synchronous agreement on $\tilde{X}$, which counts stability once per child and ignores the repaired states. So a round can lower the program's objective, and the alternation is not a descent. A descent would refit F on the program's own terms given the tree and repaired states: changes explained on the tree's edges, plus γ per unstable (state, gene).

6 Size

FUNCTIONS needs $O(N \sum_j |I_j|) = O(Nn|R|)$ variables and constraints. The agreement and instability terms are $O(Nn)$. With $\bar{k}$ the mean of $|P(c)|$, TREE has $O(N\bar{k})$ variables and constraints and TRANSITIONS has $O(N\bar{k}n)$ constraints. In total the program has $O(Nn(|R| + \bar{k}))$ variables and constraints, against $O(N^2 n + Nn|R|)$ with all cells as candidate predecessors. For all 2,167 cells ($n = 42$, $|R| = 31$, $k = 20$, $\bar{k} = 31$) the transition constraints number about 5.6×10^6 and the function representation about 1.4×10^7; the LMPP branch alone is about four times smaller.

7 Genes and prior knowledge

The processed file has 48 genes. We drop the housekeeping genes *Ubc*, *Polr2a* and *Eif2b1*, which are the normalizers, and *Egfl7*, *Gfi1* and *Sfpi1*, which Hamey et al. removed for technical reasons, leaving $n = 42$.

The regulators R are the 31 genes Hamey et al. modelled as transcription factors. The other 11 (*Cdkn2a*, *Csf1r*, *Dnmt3a*, *Itga2b*, *Kit*, *Mecom*, *Mpl*, *Procr*, *Sh2b3*, *Tet2*, *Vwf*) are mostly surface markers, signalling and epigenetic genes, and are targets only; they still inform orientation as read-outs of the cell state. This restriction is the only structural prior.

A PKN is optional. Natural candidates for these genes are ChIP-seq binding of blood transcription factors in haematopoietic progenitor cell lines, collected in the CODEX database of the same group, and the curated Boolean model of Bonzanni et al. (2013), which covers 11 of these factors in blood stem cells. Hamey et al.'s own partial-correlation network is computed from the LMPP expression data, so it is a summary of the data rather than independent prior knowledge.